openSUSE Asia Summit 2026
Yogyakarta, October 3 - 4

Building Resilient Networks with OpenMPTCProuter for Disaster Recovery

Yogyakarta, October 2026

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Bayu Putra Herlambang

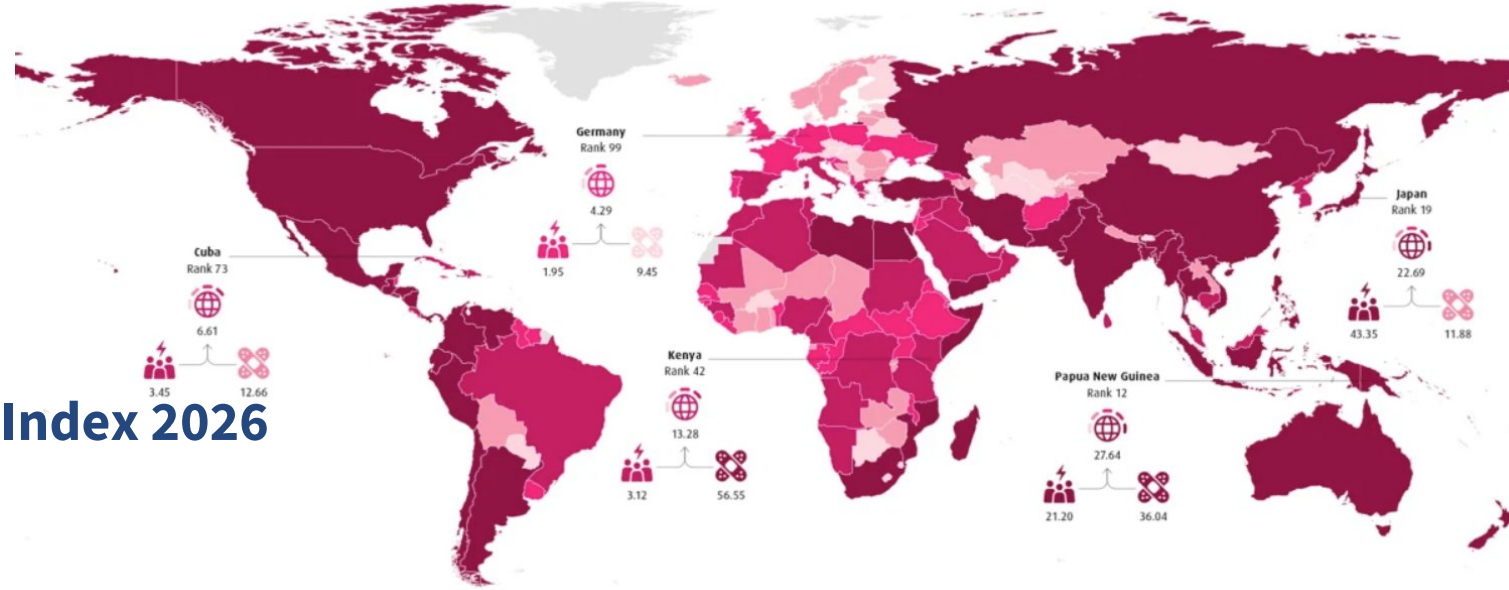
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Disaster risk in Indonesia

#3

WorldRiskIndex 2026



WORLDRISKINDEX			EXPOSURE			VULNERABILITY		
very low	0.00	2.07	very low	0.00	0.22	very low	0.00	10.09
low	2.08	3.35	low	0.23	0.74	low	10.10	15.97
medium	3.36	6.31	medium	0.75	2.13	medium	15.98	23.40
high	6.32	13.72	high	2.14	8.76	high	23.41	32.89
very high	13.73	100.00	very high	8.77	100.00	very high	32.90	100.00
no data			no data			no data		

Since 2022, fixed thresholds have been used to classify countries for the WorldRiskIndex and its components, enabling analyses of medium- and long-term trends. For each dimension and for the WorldRiskIndex, the medians of the quintile boundaries are calculated from the results of the past 20 years. Data sources: IFHV's own calculations based on CRiSIS, EMDAT, IAU, GFERR, INWE, IDMC, JRC, IMF, ILO, UCDP, UNESCO, UNHCR, UNDRR, WHO, World Bank, WorldPop, WID; additional information at https://weltrisikobericht.de/worldriskreport

THE 10 COUNTRIES WITH HIGHEST RISK		THE 10 COUNTRIES WITH HIGHEST EXPOSURE		THE 10 COUNTRIES WITH HIGHEST VULNERABILITY	
1 Philippines	46.69	1 China	71.06	1 Somalia	71.00
2 India	44.97	2 Mexico	50.69	2 Central African Republic	70.89
3 Indonesia	44.63	3 Japan	43.35	3 Chad	69.34
4 Colombia	39.80	4 India	40.83	4 Sudan	69.03
5 Mexico	39.18	5 Philippines	40.76	5 South Sudan	68.66
6 Myanmar	37.51	6 Indonesia	39.78	6 Democratic Republic of Congo	68.53
7 Mozambique	30.92	6 United States of America	39.78	7 Yemen	67.75
8 Russian Federation	30.58	8 Australia	32.11	8 Afghanistan	67.36
9 China	29.54	9 Colombia	31.59	9 Niger	66.63
10 Somalia	29.12	10 Russian Federation	28.37	10 Ethiopia	66.33

Communication in network outages

Impacted infrastructure

Base stations (BTS) can overload and fiber links can be severed

Command post operations must continue

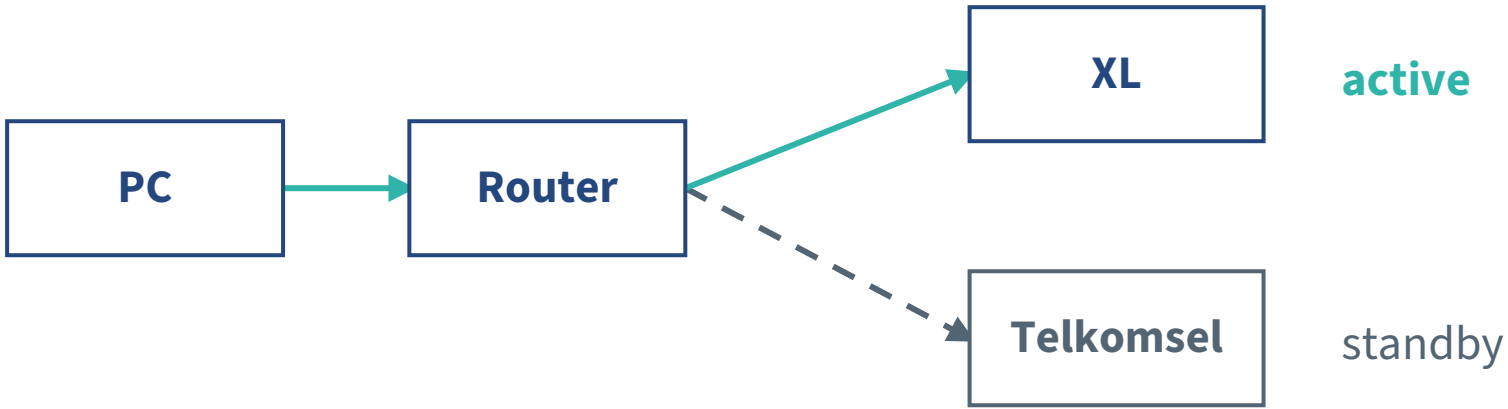
Prioritize coordination messages, report submissions, and calls

Remaining opportunities

Utilize multiple surviving links and test their continuity.

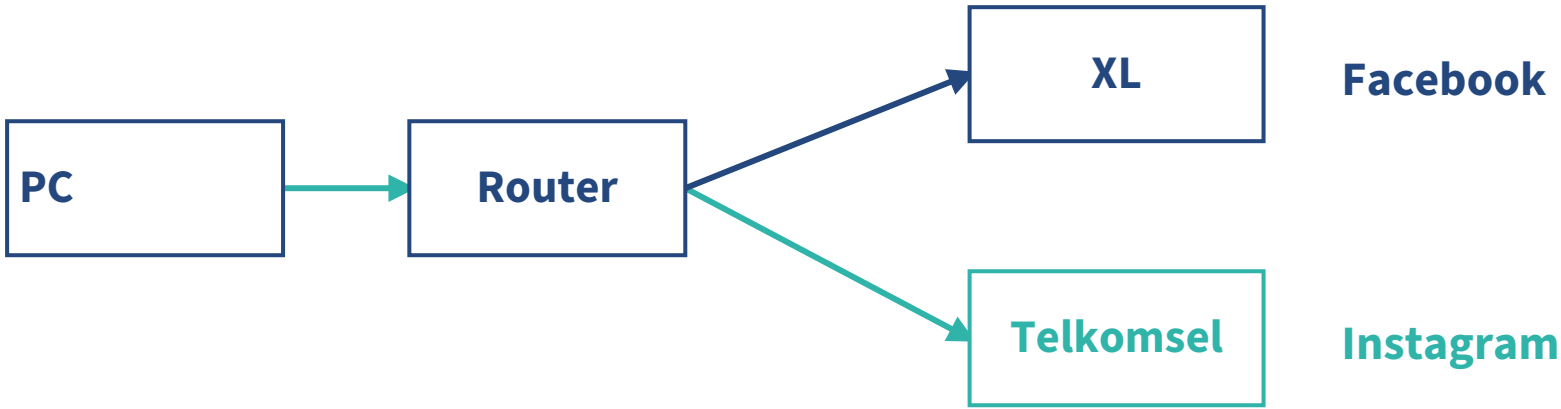
Failover

Only 1 link is active; the others remain on standby



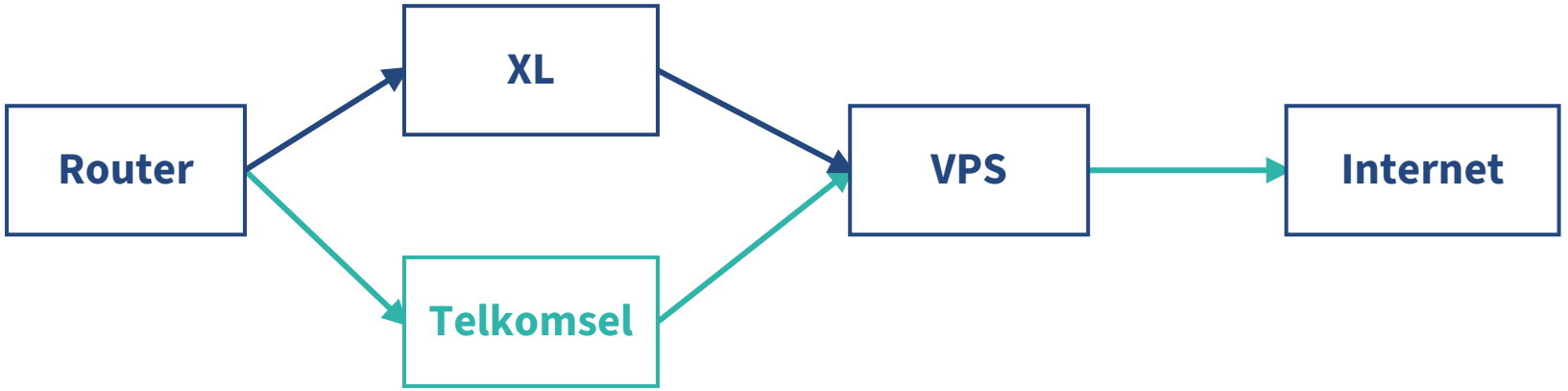
Load balancing

Distributing connections across multiple links



Aggregation with OpenMPTCProuter

Multiple links connect the OMR router to the VPS.



Key differences

Method	Type	Concern
Failover	Main & Backup	Switchover time
Load balancing	Distributed connections	Single flow typically uses one link
OMR aggregation	Multiple links to VPS	Link quality and endpoint capacity

Emergency Network Needs

- Seamless failover: Connection must not drop when one upstream link loses signal
- Bandwidth aggregation: Capable of combining weak signals into a single stable link
- Portable & power-efficient: Hardware must be compact and support power from a battery, power bank, or solar panel
- Security: Transmitted data must be encrypted

What is OpenMPTCProuter?

An OpenWrt-based open-source solution for aggregating multiple internet connections.

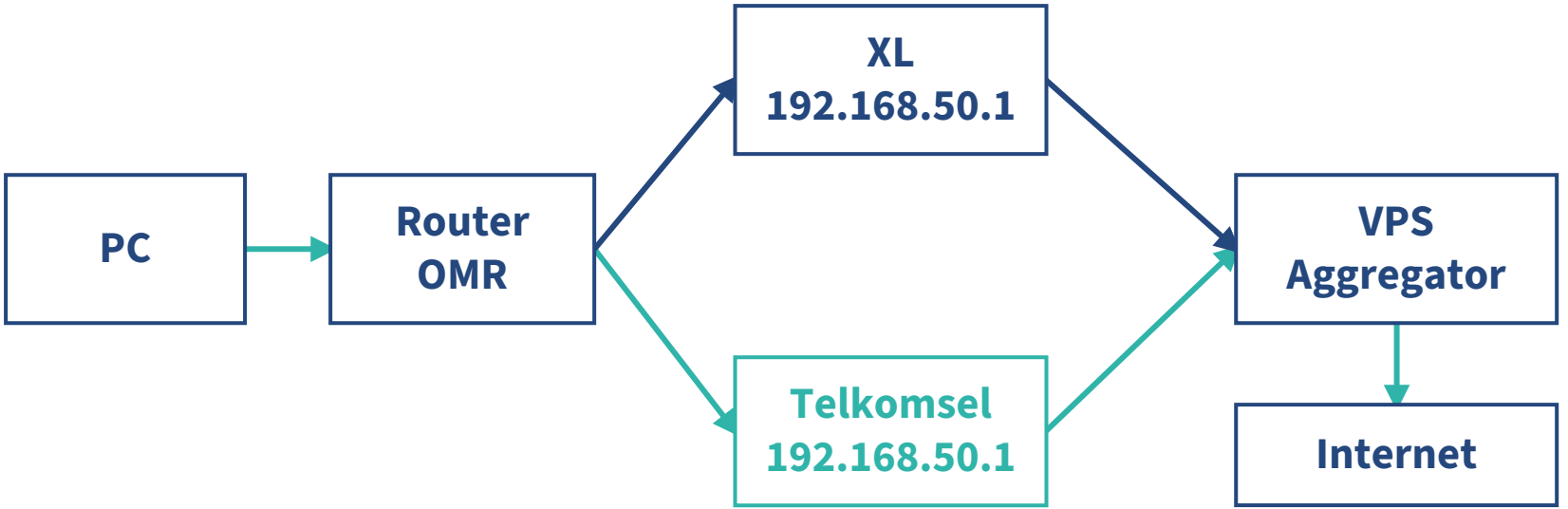
Field router

Manages WAN interfaces, MPTCP, and proxy or VPN.

Off-site VPS

Serves as the tunnel endpoint and internet traffic exit point.

Topology


PC
Router
OMR
XL
192.168.50.1
Telkomsel
192.168.50.1
VPS
Aggregator
Internet

Hardware

Component	Function / requirement
OMR Router	Raspberry Pi 3B+ or higher
Modem + SIM	Internet source
VPS	Public IP, sufficient capacity and quota

Limitations and field resilience

Shared failure domains

Two SIM cards may share the same tower, backhaul, or power source.

VPS dependency

If the endpoint is unreachable, this aggregation path is also affected.

WAN quality

Latency, packet loss, and device capacity determine the outcome.

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Hands-on

1. Flash router image

Download the image for your device model. Extract to a .img file.

```
lsblk -o NAME,SIZE,MODEL,MOUNTPOINTS

sudo dd if=omr.img of=/dev/sdX \
  bs=4M status=progress conv=fsync
```

2. Prepare the VPS

```
wget https://www.openmptcprouter.com/\
server/debian-x86_64.sh

less debian-x86_64.sh
sudo sh debian-x86_64.sh
```

3. Configure WAN and LAN subnets

Segment	IP
WAN 1	Telkomsel Modem 192.168.50.0/24
WAN 2	XL Modem 192.168.60.0/24
LAN	OMR 192.168.100.1

4. Connect OMR to the VPS

Access router from LAN

Open 192.168.100.1 and set the admin password.

Open the OMR wizard

Go to System / OpenMPTCProuter / Wizard. Enter the server IP and key.

Configure WAN

Enter the IP address and gateway for each WAN interface.

5. Verification

Each WAN link is functional

Test the gateway and internet access of each link separately.

Router is connected to the VPS

Check the OMR status page, tunnel state, and traffic across both WANs.

Conclusion

Aggregation provides additional path options

OMR bonds multiple WAN links to a VPS endpoint.

Resilience must be verified

Measure session disruption during WAN failure and link recovery.

Field readiness remains decisive

Prepare independent links, power supply, and operational procedures.

Guides and references

OpenMPTCProuter: router installation

OpenMPTCProuter: VPS installation

OpenMPTCProuter: WAN and server configuration

OpenMPTCProuter: technologies and testing

WorldRiskReport 2025

Links are clickable. Use documentation matching your software version.

Thank you



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